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Department of Information and Communication Technology

COMPUTER VISION FINAL PROJECT

Topic:

Colonoscopy Polyp Segmentation

Group 47

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Contents

Abstract 3
1. Introduction 3
1.1 Context & Motivation 3
1.2 Related Works 4
1.3 Objectives 5
2. AI Methodology 5
2.1 Dataset overview and preprocessing 6
2.1.1 Raw dataset 6
2.1.2 Data preprocessing 7
2.2 Model Selection 8
2.3 Segmentation pipeline 9
3. Results 10
3.1 Model evaluation 10
3.1.1 Quantitative evaluation 10
3.1.2 Qualitative evaluation 12
3.2 Polyp Segmentation and Risk Assessment System 14
4. Conclusion 16
5. Future work 17
Reference 17

Abstract

In this project, we focus on helping doctors find and evaluate colon polyps more easily using artificial intelligence. While many AI models are very smart, they are often too hard for doctors to use in real life. We built a system that combines a strong deep learning model called UNet++ with a simple web interface. We used a pre-trained ResNet34 encoder to help the model learn quickly from medical images, even when we did not have a huge amount of data. Beyond just finding polyps, we added a special feature that colors the results red or green to show the level of risk based on the polyp's size. Our results show that the model is very accurate, and by putting it on a web platform, we have created a practical tool that can act as a second pair of eyes for medical professionals.

1. Introduction

1.1 Context & Motivation

Colorectal cancer (CRC) is a major global health issue and one of the most common causes of cancer-related deaths. However, it rarely appears overnight. The disease typically begins as benign polyps—small clumps of cells that form on the inner lining of the colon or rectum. Over time, some of these polyps mutate and become cancerous. Because we know exactly how this progression works, doctors have a clear window of opportunity: if we can find and remove these precancerous polyps early, we can effectively stop cancer before it starts.

The standard way to do this is through a colonoscopy. During this procedure, a doctor uses a flexible camera to visually examine the inside of the colon, looking for any abnormalities. If they spot a polyp, they can usually remove it right then and there. But while colonoscopies save lives, the process is far from perfect because it relies entirely on human observation. Endoscopists must navigate a camera through a complex, moving environment while staring at a monitor, often for hours on end. Polyps can be incredibly hard to see—many are flat, blend in with the surrounding tissue, or hide behind folds in the colon. Throw in poor lighting, camera blur, and simple eye fatigue, and it’s no surprise that human error is a significant factor. In fact, clinical studies show that doctors can miss up to 25% of precancerous polyps during routine screenings.[1]

This high miss rate is exactly why there is a massive push to bring computer vision into the operating room. We need a system that acts like a tireless second pair of eyes. This project focuses specifically on semantic segmentation rather than just drawing a basic box around a polyp. In a clinical setting,

knowing the exact boundaries of a lesion pixel-by-pixel is critical. It helps doctors assess the size of the polyp, understand its shape, and plan how to safely remove the entire thing without leaving dangerous tissue behind. By building an AI that can automatically and precisely highlight these regions in real-time, we can reduce the mental strain on doctors, make colonoscopies more consistent, and ultimately help prevent more cases of colorectal cancer.

1.2 Related Works

In this section, we review previous research directions in colorectal polyp detection and segmentation. We categorize these works into two main approaches: traditional feature-based methods and modern deep learning architectures. Finally, we discuss how our work addresses the limitations of these studies by focusing on clinical accessibility and real-time visualization.

Regarding traditional feature-extraction approaches, early attempts in polyp recognition relied heavily on manual feature engineering. Researchers like Tajbakhsh et al focused on identifying specific visual clues, such as the distinct reddish hues of polyps or the characteristic curvature of their edges and then processed these data points through classic classifiers like Support Vector Machines (SVM). [2] However, the internal environment of the colon proved too chaotic for such methods as medical instruments, fluid reflections, and inconsistent lighting frequently confuses these rigid algorithms. Because polyps exhibit immense variety in shape and texture, these older methods generated high rates of false positives and struggled to generalize across diverse clinical datasets.

The field experienced a significant shift with the emergence of deep learning and encoder-decoder architecture, specifically through Convolutional Neural Networks (CNNs). The introduction of UNet architecture marked a turning point for medical imaging by using a symmetrical structure to pinpoint objects at the pixel level. [3] Subsequent advancements have built extensively on this foundation by incorporating Transfer Learning, where powerful encoders like ResNet or EfficientNet—pre-trained on millions of everyday objects—are integrated into the network to provide a superior head start in feature extraction. Modern frameworks have further streamlined this process, allowing researchers to easily snap these advanced encoders into U-Net structures.

However, looking at clinical visualization and alert systems in recent research, much of the community remains focused on raw mathematical accuracy while ignoring the practical needs of medical practitioners. A purely theoretical model lacks the necessary interface for real-world utility, which is why bridging the gap between high-performance neural networks and an accessible, intuitive user interface is a central focus of this project. Unlike previous works that stop generating binary masks, our approach emphasizes a color-coded risk assessment system to provide immediate, actionable feedback for endoscopists.

1.3 Objectives

The primary goal of this project is to develop a reliable, deep learning-based system that can accurately segment colon polyps and encapsulate them within a platform that ensures clinical accessibility. To achieve this, our first objective, Architectural Design and Transfer Learning Integration, focuses on building a robust semantic segmentation framework using the PyTorch ecosystem. By leveraging the UNet++ architecture combined with a pre-trained ResNet34 encoder, we aim to ensure the system can accurately extract intricate morphological features from messy endoscopic environments, even when training data is limited.

Building upon this technical foundation, our second objective involves Rigorous Optimization and Clinical Validation. We aim to apply intensive data augmentation to simulate the unpredictable conditions of real-world colonoscopies and evaluate the model using stringent metrics, specifically the Dice Similarity Coefficient and Intersection over Union. This ensures that the AI's predicted boundaries are not just statistically high, but also geometrically precise when compared to expert medical annotations, establishing a reliable "Best Model" for deployment.

Finally, we address the Development of a Practical Interface and Risk Assessment System. This objective seeks to transform the mathematical success of the model into a functional tool via the Gradio framework. Beyond simple segmentation, this phase focuses on developing a specialized logic that calculates the related area of identified lesions and provides immediate visual feedback through a red and green color-coding system. By doing so, we aim to provide physicians with an intuitive "second pair of eyes" that simplifies the interpretation of AI data into immediate clinical alerts.

2. AI Methodology

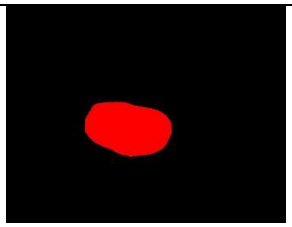
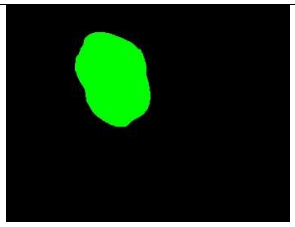
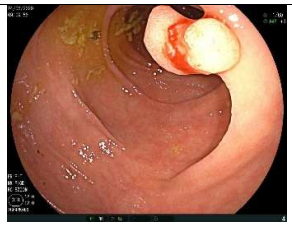
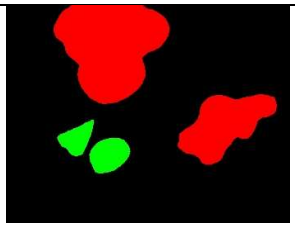
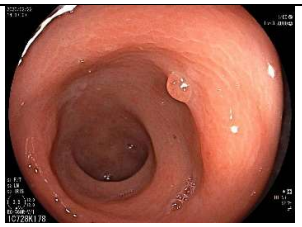
To realize the goal of building an automated system for identifying and warning of colorectal tumor risks, the project requires the in-depth application of Artificial Intelligence (AI) algorithms in the field of Computer Vision. In this chapter, we will present in detail the AI methodology, starting from the analysis and preprocessing of image datasets, followed by the selection and training of a suitable model, thereby obtaining the system's brain before integration into the system.

2.1 Dataset overview and preprocessing

The performance of any AI model depends heavily on the quality of the data it uses for learning. Therefore, it is necessary to understand and prepare the input data. First, we will explore the origin and characteristics of the raw endoscopic images.

2.1.1 Raw dataset

The raw data used in this study were extracted from the "BKAI-IGH Neo Polyp" dataset on the Kaggle platform: https://www.kaggle.com/competitions/bkai-igh-neopolyp/data.[4] This is a specialized medical dataset, collected from September 30, 2021, to January 1, 2025, through a collaboration between experts at BKAI and the Institute for Hepatobiliary Research (IGH). The structure and distribution of the dataset are illustrated in detail on the table below:

train	train_gt	test
		
		
		
Total: 1000	Total: 1000	Total: 200

The data is divided into three main sets: the train set (containing 1000 raw endoscopic images), the train_gt set (containing 1000 corresponding Ground-truth masks for training), and the test set (containing 200 unlabeled images for evaluating the real-world model).

Specifically, the images in train_gt are not simply binary (black and white) masks but also include color to convey clinical classification meanings for physicians. Red indicates neoplastic polyps – tumors with a risk of developing into cancer (as illustrated in row 1 of Table 2.1). Green indicates non-neoplastic polyps – benign tumors with low risk (as illustrated in row 2 of Table 2.1). Black indicates healthy mucosal tissue or the endoscopic background.

Endoscopic procedures can be very complex, as illustrated in the third row of Table 2.1, where a single frame contains multiple tumors with varying degrees of risk (both red and blue). Differentiating these polyp types visually is extremely difficult because identifying features such as surface structure or vascular patterns is often inaccurate, and the boundary between benign and potentially malignant tumors is very thin. Therefore, manual identification is not only time-consuming but also carries a high risk of error. This is what prompted us to develop an AI model to support the automation of this diagnostic process.

2.1.2 Data preprocessing

In the raw dataset, the original endoscopic images exhibit significant differences in resolution, lighting conditions, and noise. Directly feeding this heterogeneous data into a deep learning model would hinder feature extraction and reduce training stability. Therefore, these raw images must be pre-processed before proceeding further.

The images need to be resized and normalized before being fed into the model. We need to resize the images to the standard size of 352 x 352 pixels because the original endoscopic images are often very high-resolution and inconsistent, and this allows the model to compute faster while retaining all the important details of the tumor. If kept at the original size, the computer would be unable to process them due to RAM overload. At the same time, the pixel color values are brought to the range [0, 1] to help the optimization algorithm operate more stably and converge faster during training.

In addition, a major challenge when using the original dataset is that the test set completely lacks ground truth. This makes it impossible to calculate evaluation parameters and visualize results for comparison. To address this, we took all 1000 images from the two training sets and split them to an 8:2 ratio, with 80% of the data used for model training and the remaining 20% for calculating evaluation parameters and visualization. Images from the original test set will be used for testing on the system.

2.2 Model Selection

After preprocessing, the data are ready for deep learning analysis. The next crucial stage in our method is selecting a robust model capable of extracting complex morphological features from these preprocessed inputs.

Our input data consists of colonoscopy images containing tumors of various shapes, sizes, and colors, making segmentation extremely complex. Therefore, several deep learning models could be used: traditional convolutional neural networks (CNNs) (UNet [3]), improved CNNs (UNet++ [5]), or modern Vision Transformer models (ColonFormer [6], TransUNet [7]).

Model	Architecture Type	Keeps Edge Details	Sees the Whole Image	Computing Power Needed	Good for Web Apps
UNet [3]	Basic CNN	Fine	Limited	Low	Yes
UNet++ [5]	Upgraded CNN	High	Limited	Medium	Yes
TransUNet [7] / ColonFormer [6]	Vision Transformer	High	Excellent	Very High	No

Based on this table, we chose the UNet++ [5] model combined with a ResNet34 [8] encoder. We made this choice to balance high accuracy with the ability to run smoothly on a website.

Modern Transformer models like TransUNet [7] and ColonFormer [6] are excellent at seeing the big picture of the whole image [6]. However, they are very heavy and need a massive amount of computing power—up to 105.5 million parameters [7]. Because they are so heavy, they run too slowly and are not suitable for a real-time, lightweight web application.

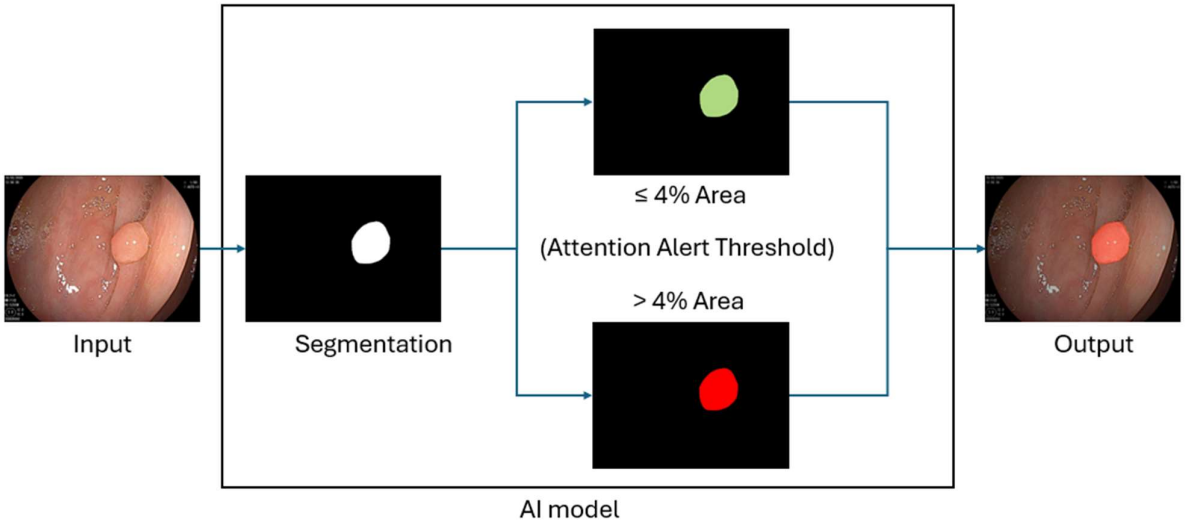
On the other hand, UNet++ gives us the best balance between speed and accuracy. In medical images, the border between a polyp and normal tissue is often very blurry. UNet++ uses special internal connections (nested bypass connections) to keep the tiny details at the edges of the tumor, which basic UNet often loses [5]. Also, by pairing it with the ResNet34 encoder, our model stays light enough to process images quickly without making the website lag.

We know that UNet++ has some weak points: it does not see the whole image as well as Transformers [6], and it can be sensitive to image noise. However, we have fixed these issues by carefully resizing and standardizing (normalizing) the images before feeding them to the AI, as

explained in the previous section. In the end, its ability to find clear polyp edges and run quickly makes UNet++ the best practical choice for our system..

2.3 Segmentation pipeline

After selecting a suitable AI model, the final step in our methodology is to present the entire end-to-end segmentation process to transform raw endoscopic images into color-coded visualizations, as shown in the diagram below.



The process is tightly structured into the following processing stages: Input and preprocessing, Segmentation and Risk Coloring by AI model (UNet++), and Output.

The process begins when the doctor uploads the raw endoscopic images into the system. Users don't need to worry about inconsistencies in resolution or lighting conditions across different devices, as the system automatically performs background normalization steps to ensure optimal image quality before analysis, effectively eliminating hardware-induced aberrations.

After receiving the processed images, the UNet++ AI model scans the entire frame, detecting and accurately identifying the boundaries of polyps hidden within the colon mucosa, even those with complex morphologies. It immediately calculates the relative size of the tumor on the screen and assigns a color code overlay to the original image. Specifically, a red overlay is automatically assigned to large tumors, occupying more than 4% of the frame area. Conversely, a green overlay is assigned to smaller polyps, occupying 4% or less of the frame area. The result provided to the user is a comprehensive visual diagnostic image, clearly displaying the location and severity of the tumor in just seconds.

However, in actual endoscopy, the camera continuously zooms in and out. Therefore, a small, benign 2mm polyp photographed at close range might be flagged with a red alert due to its relatively large screen area, while a large, dangerous tumor viewed from a distance might only trigger a green alert. Thus, the current system primarily acts as an attention filter to prevent missed lesions rather than a definitive histological diagnosis to differentiate between neoplastic and non-neoplastic tissue. The final diagnosis of the tumor's nature still requires the physician's clinical expertise in observing surface microstructure and vascular patterns, thus paving the way for the integration of specialized AI classification networks in future developments.

3. Results

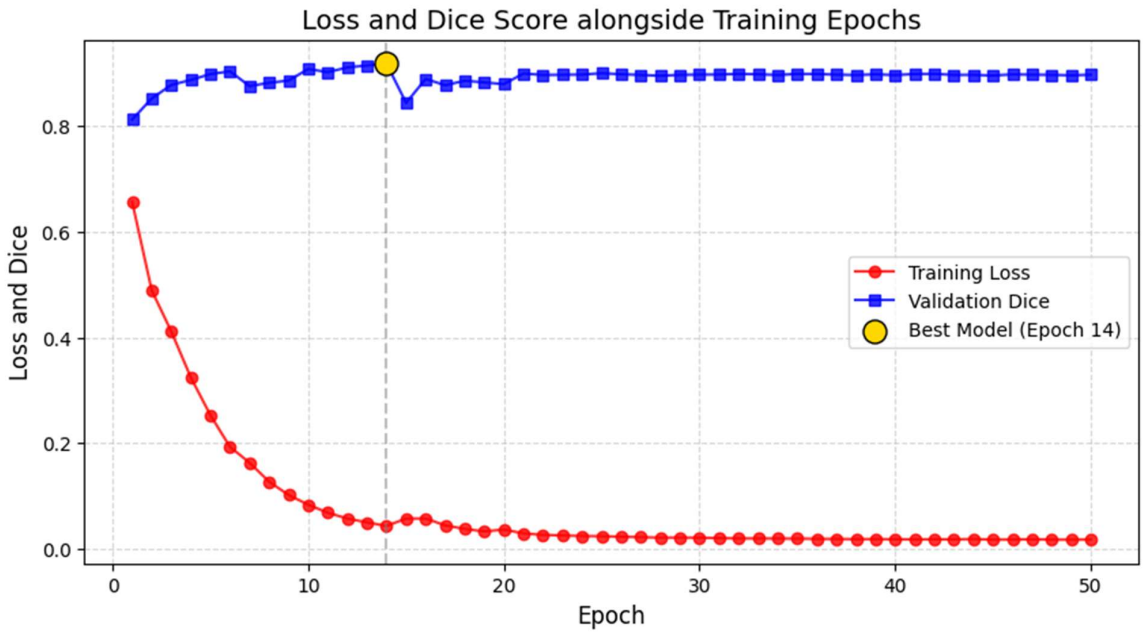
Having developed the UNet++ [5] AI model in the previous chapter, we now need to verify its performance. We evaluate the model's predictive capabilities through both quantitative training metrics and qualitative visual results. This will allow us to confidently apply it to the Polyp Segmentation and Risk Assessment System.

3.1 Model evaluation

We begin with an objective analysis using standard mathematical indicators to quantify the accuracy of the model.

3.1.1 Quantitative evaluation

The graph "Loss and Dice Scores along with Training Epoch Number" below provides a visual overview of the learning speed and stability of our UNet++ [5] architecture using the ResNet34 [8] encoder.



The core objective of the training process is to minimize the discrepancy between the model's predicted polyp mask and the actual Ground Truth data annotated by medical professionals. To comprehensively monitor this learning process, we set a training limit of 50 epochs. This specific duration was chosen because our architecture leverages the ResNet34 encoder, which already possesses robust feature extraction capabilities from ImageNet. Therefore, the network only requires a moderate number of cycles to adapt to the specific endoscopic data; extending the training process further would incur unnecessary computational costs and increase the risk of overfitting.

Throughout these 50 training cycles, we relied on a combined loss function to optimize the model, which is mathematically defined as $Loss = 0.5 * L_{BCE} + 0.5 * L_{Dice}$.

In this formulation, L_{BCE} (Binary Cross-Entropy Loss) evaluates the pixel-level classification accuracy by penalizing incorrect individual pixel predictions, while L_{Dice} (Dice Loss) enforces global spatial consistency by maximizing the geometric overlap between the predicted and actual polyp boundaries.

Analyzing the training loss curve (red), the model adapts to the endoscopic data extremely quickly. From the very first epoch, the loss value drops sharply from a high to below 0.2. This rapid decrease demonstrates the efficiency of the ResNet34 encoder, the model quickly identifies basic tumor structures without needing to relearn from scratch. After about 10 epochs, the loss curve approaches 0 and maintains a stable state with minimal fluctuations. This smooth convergence confirms that our combined loss function effectively minimizes training instability, such as the gradient bursts often seen in segmentation tasks.

Simultaneously, the system was evaluated using the Validation Dice Score (blue curve). The Dice Score [9] is a harmonic assessment of spatial overlap, calculated as $Dice = \frac{2*TP}{2*TP+FP+FN}$.

In that formula, TP (True Positive) represents correctly identified polyp pixels; FP (False Positive) represents healthy backgrounds incorrectly marked as polyps (false alarm); and FN (False Negative) represents areas of polyps missed. This curve increases inversely with Loss, showing a continuous improvement in generalization ability. Most notably, the optimal balance is achieved at Epoch 14 (marked in yellow), where the model recorded the highest Validation Dice score of 0.9175. At this point, the model achieved a perfect balance between learning features and generalizing them to unseen data. Therefore, the weights at Epoch 14 were extracted and retained as the "Best Model" for later practical implementation, preventing the slight overfitting observed in subsequent epochs.

Beyond the primary Dice score, we further quantified the performance of this Best Model using three additional critical metrics:

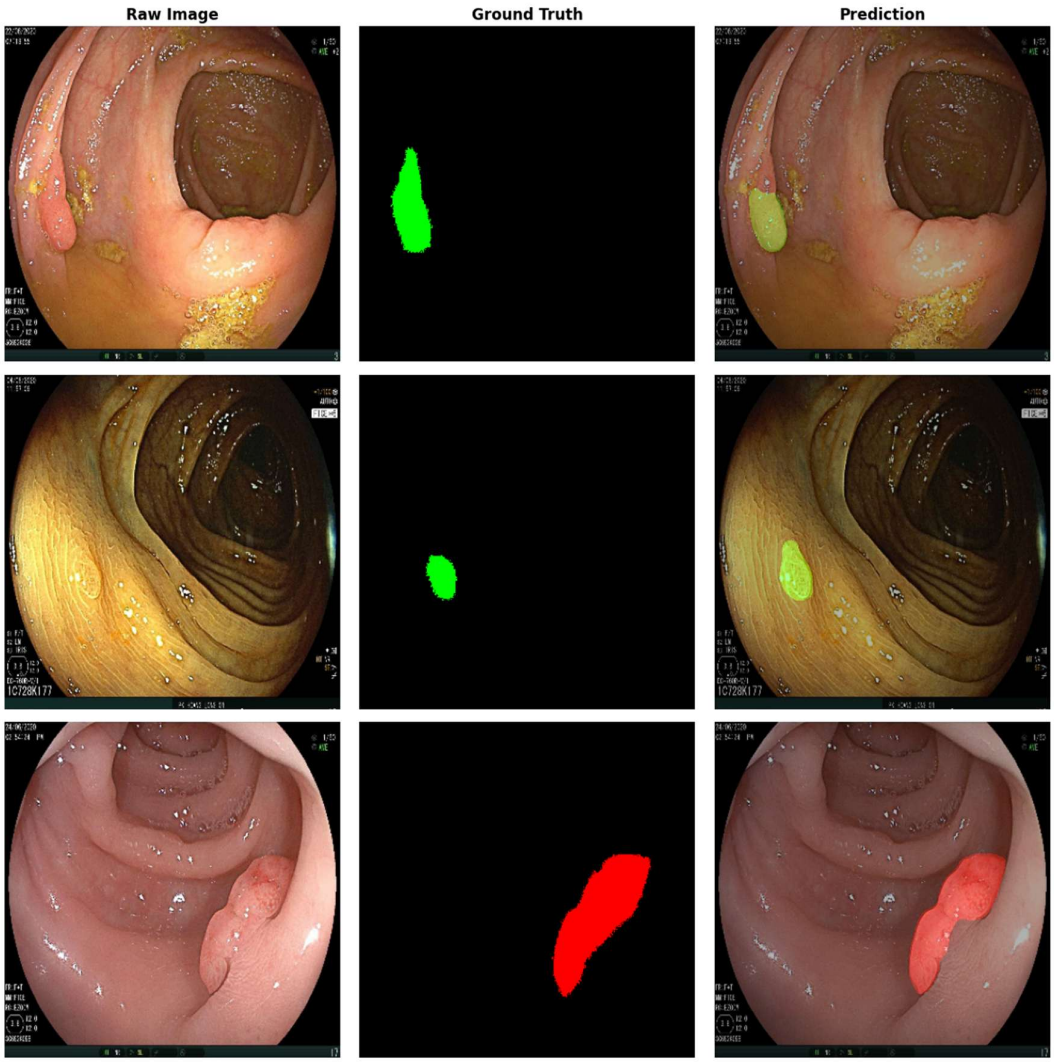
IoU (Intersection over Union) [10] $= \frac{TP}{(TP+FP)+(TP+F\quad)-TP} = \frac{TP}{TP+FP+FN} = 0.8497$ represents the spatial overlap, measures how perfectly the predicted mask aligns with the actual tumor. A score of nearly 85% ensures that the AI's predicted boundaries tightly wrap the true geometric shape of the polyp without bleeding into the healthy background.

$Precision = \frac{TP}{TP+FP} = 0.9417$ measures the reliability of the model's positive predictions. A score exceeding 94% indicates that when the system highlights a region, it is almost certainly a true polyp, thereby minimizing false surgical alarms.

$Recall = \frac{TP}{TP+} = 0.8982$ represents the model's ability to detect all existing tumors. In medical diagnostics, a high Recall is paramount to ensuring that dangerous neoplastic polyps are not overlooked and left untreated.

3.1.2 Qualitative evaluation

While quantitative metrics provide a mathematical measure of accuracy, qualitative evaluation is essential for visually verifying the model's behavior in real-world clinical scenarios. The table below presents a side-by-side visual comparison of three representative experimental cases.



This table has three columns: Raw Image (initial endoscopic input), Ground Truth (accurate polyp region), and Prediction (final diagnostic result generated by our system).

Over the table, the UNet++ model demonstrates good accuracy. In all three examples, the predicted overlays closely match the complex and irregular morphological boundaries of the actual polyps shown in Ground Truth. The AI model successfully differentiated tumor tissue from the surrounding healthy colonic mucosa, demonstrating robustness in the face of common endoscopic imaging challenges such as light reflection (glare from camera lights), lack of clear contrast, and varying tissue textures.

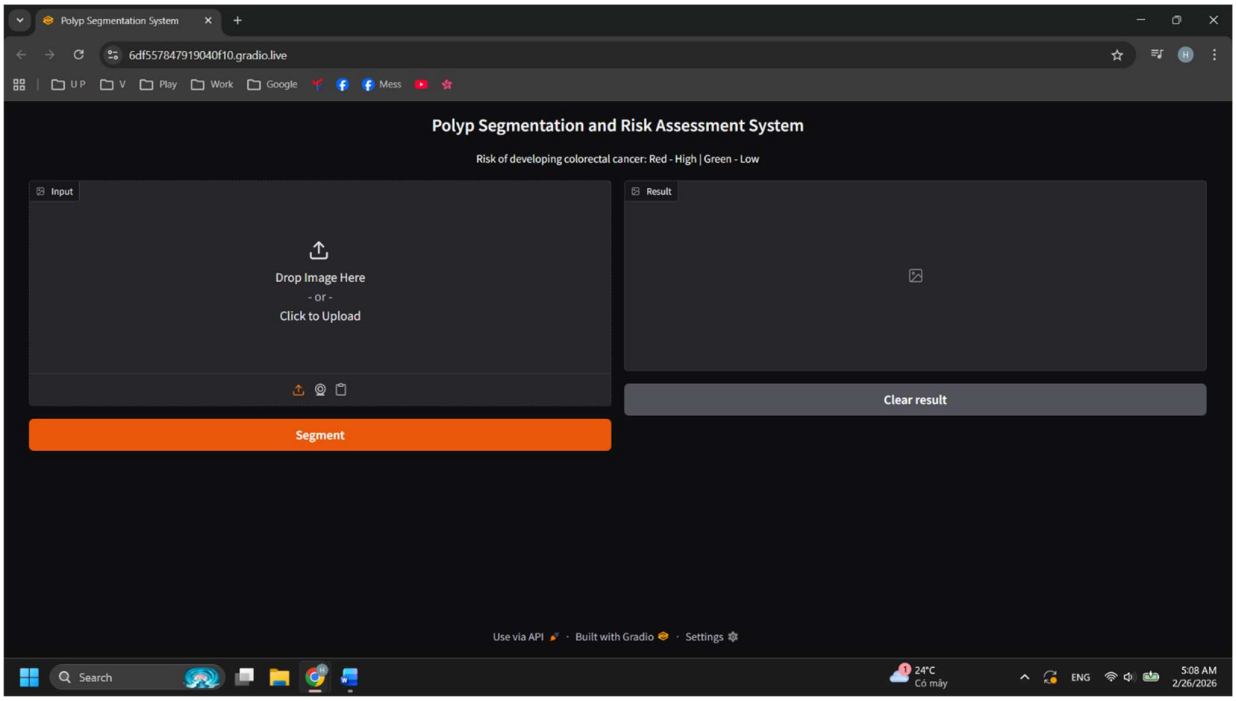
Furthermore, the visual results demonstrate the practical effectiveness of our Attention Alert Threshold logic. In the first two rows, the detected polyps were relatively small, occupying 4% or less of the total frame area. By design, the system accurately assessed their relative size and assigned

a green overlay to these lesions, indicating a lower immediate visual priority. Conversely, the third row featured a prominent, elongated tumor that clearly exceeded the geometric threshold. The system triggered a high attention alert, applying a red overlay to immediately draw the physician's attention to this large anomaly.

Overall, these qualitative visualizations strongly reinforce the high Dice scores and IoU achieved during the quantitative assessment. This demonstrates that the system successfully translated statistical accuracy into highly reliable, intuitive, and user-friendly visualizations.

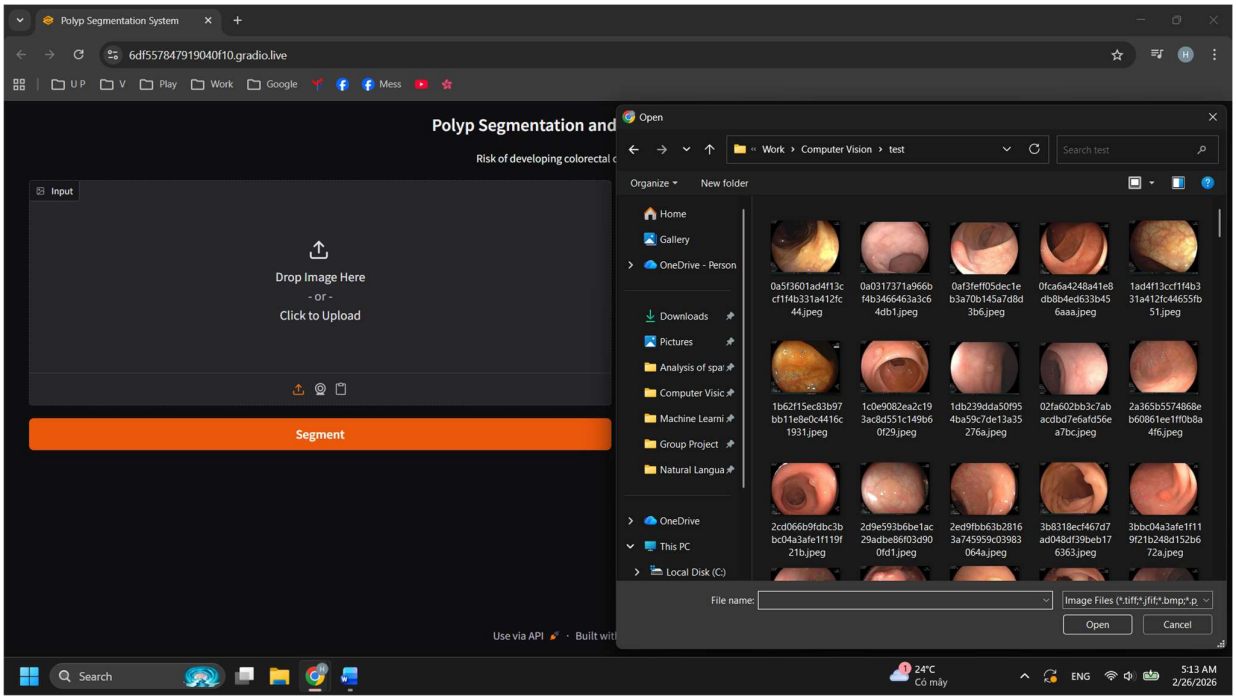
3.2 Polyp Segmentation and Risk Assessment System

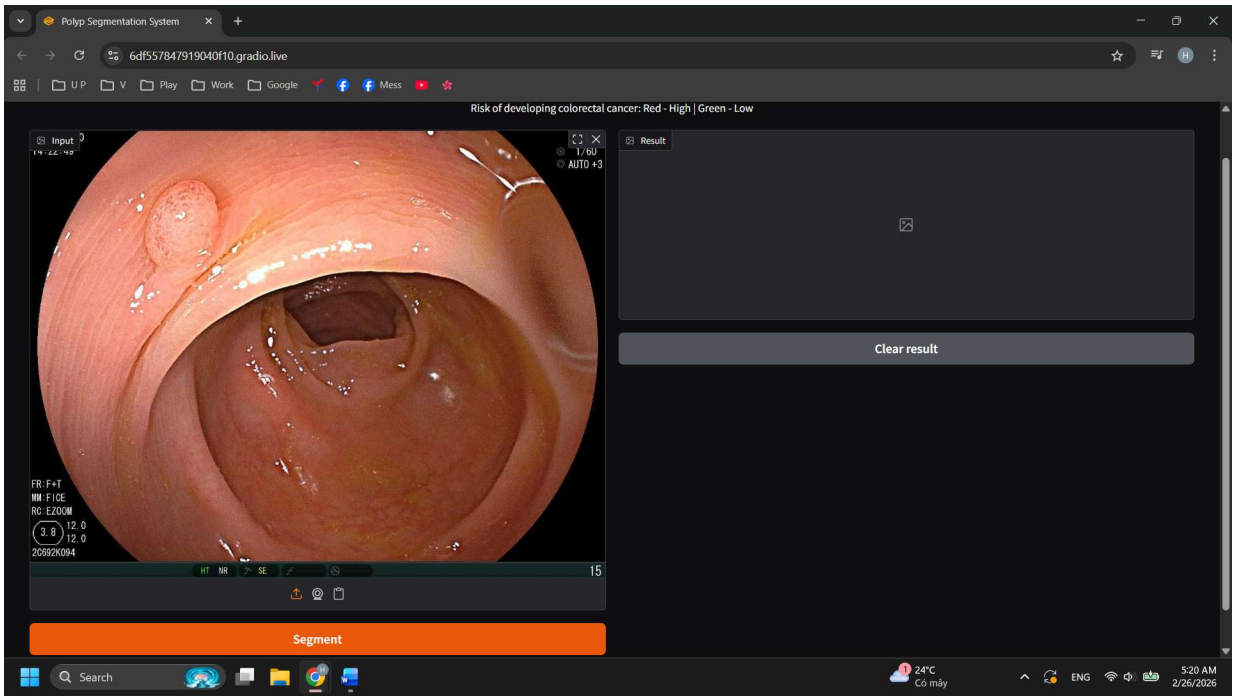
With the promising results we obtained in the model evaluation, we can now confidently use our UNet++ AI model to apply to the Polyp Segmentation and Risk Assessment System. In this system, the AI model has been encapsulated in an intuitive user interface, allowing users to easily utilize it.



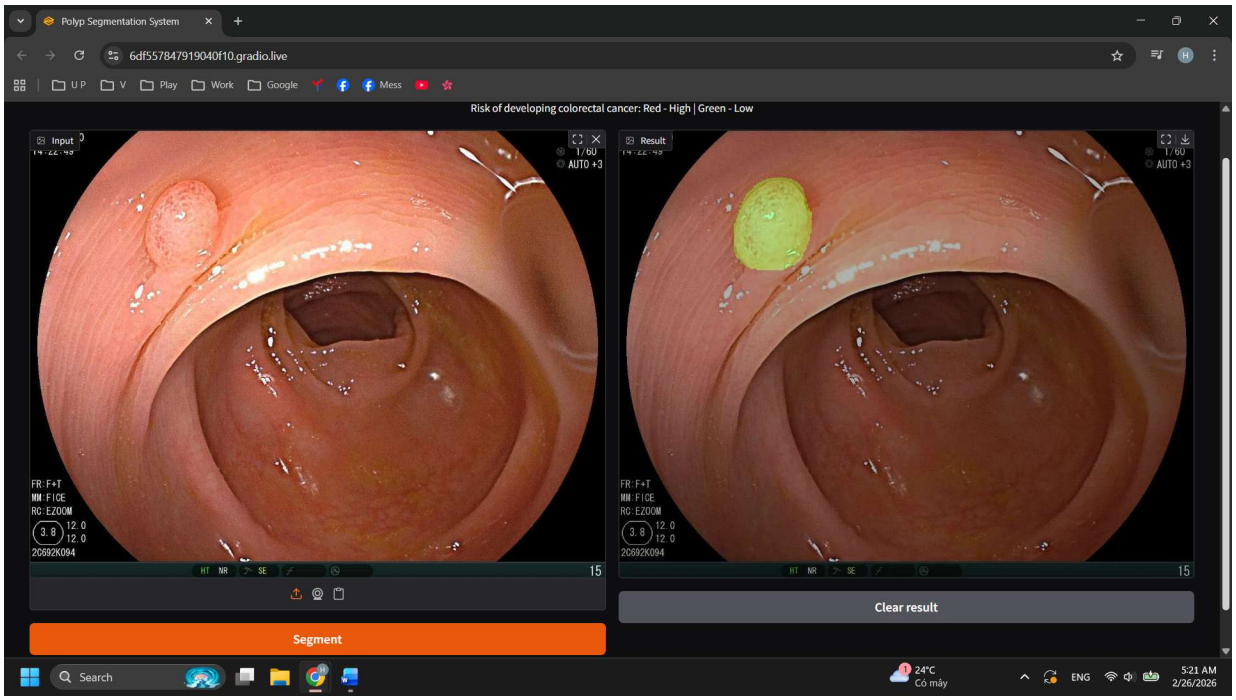
The image above illustrates the initial user interface of the minimalist web application, designed with a dark theme layout and developed using the Gradio [10] framework. The interface is visually divided into two main sections: the "Input" panel on the left, allowing users to drag and drop or upload raw endoscopic images, and the "Results" panel on the right for the AI model's output. A prominent orange "Segmentation" button serves as the primary trigger to initiate automated analysis. Furthermore, a clear annotation is consistently displayed at the top (Risk of developing colorectal

cancer: Red - High | Green - Low), ensuring a simple and transparent risk stratification logic for the user before interaction.





When the user clicks "Clidk to Upload," their file library will open, allowing them to select images as input. If it's a raw image, the AI model can process it.



After the endoscopic images are loaded into the input frame, the user simply clicks the "Segment" button to activate the AI processing. Within seconds, the system will return the diagnostic results directly in the right-hand frame. Users can save their results. They can also start over by pressing the "Clear result" button.

4. Conclusion

We have successfully built a system that can find colon polyps and tell us how much attention they need. By following our project plan, we showed that using modern AI structures like UNet++ along with transfer learning is a great way to handle complex medical images. We found that our model does not just look good on paper with high scores; it performs very steadily and accurately when we test it with new data.

The part of this project we are most proud of is how we turned a complex mathematical formula into a tool that anyone can use. By creating a web app that uses red and green colors to highlight risks, we made it much easier for doctors to understand what the AI is seeing. We believe we have met our main goal of connecting advanced technology with real-world medical needs. Our system is not just a piece of code; it is a helpful assistant that can make a real difference in how doctors perform screenings.

5. Future work

Looking ahead, we have several ideas to make our system even better. First, we want to improve how the AI thinks about risk. Right now, we use the size of the polyp to set the color, but in the future, we want the AI to look at the patterns and blood vessels on the polyp's surface. This would allow the system to tell the doctor if a polyp is truly dangerous or just harmless growth.

Second, we want to make the system work with live video instead of just still photos. We plan to optimize our code so that it can process the video feed from an endoscopy camera in real-time without any lag. We also hope to collect more data from different hospitals to make sure our AI works well for everyone, no matter what kind of camera they use. By making these changes, we hope to move closer to a version of our tool that can be used every day in operating rooms to help save lives.

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